A.R.C.-6 EMERGENCY PROTOCOLS MANUAL

Six Primary Reactors, Two Auxiliary Breeder Cores, and A.R.C. Field Safeguards

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1.0 - PURPOSE

This manual defines emergency procedures for the final A.R.C.-6 configuration: six parallel primary fusion reactors, two isolated auxiliary cylindrical breeder cores, the A.R.C. Field system, cryogenic infrastructure, isotope processing, and direct-energy conversion.

2.0 - CONFIGURATION CONTROL

- Primary reactor count: six.
- Auxiliary breeder-core count: two, B1 and B2, not included in A.R.C.-6 designation.
- The central A.R.C. Field lattice / coherence spine is not a reactor.
- Any safety display reporting eight, twelve, or another primary-core count is invalid and requires configuration-control lockout.

3.0 - PRIMARY-CORE EMERGENCY SHUTDOWN

- Isolate the affected core from the six-channel fuel, heating, DEC, and power-dispatch systems.
- Execute certified fast plasma termination within the vacuum vessel.
- Maintain remaining primary reactors in parallel only if common-bus and A.R.C. Field stability margins remain valid.
- Trigger magnet energy extraction only when required by the fault.
- Do not route plasma through doors or into a remote dump chamber.

4.0 - BREEDER-CORE EVENT

- Inhibit pulses and isolate the affected cylindrical breeder bay.
- Close isotope manifolds and transfer tritium-bearing gas to monitored hold-up tanks.
- Maintain independent shielding, vacuum, and detritiation boundaries.
- Suspend both breeder cores during uncertain cross-bay isotope accountancy or shared utility faults.
- Breeder shutdown does not change the six-primary-core A.R.C.-6 count.

5.0 - A.R.C. FIELD FAILURE

Adaptive Resonance Confinement (A.R.C.) Field Technology is a classified photonic-magnetic confinement system that generates a resonance-stabilized coherence envelope for AGI substrates. The field suppresses quantum decoherence, stabilizes photonic neuromorphic pathways, and enables long-range temporal correlation across multi-substrate AGI systems. Full Observer capability requires continuous operation within an active A.R.C. Field.

Field failure is treated as a critical computation-environment fault, not automatically as a plasma-confinement failure. Full Observer capability cannot continue without the active envelope.

5.1 - Field Failure Response

- Freeze nonessential AGI self-modification and long-horizon planning.

- Preserve volatile state and transfer essential functions to hardened conventional compute partitions.
- Suspend long-range temporal-correlation processes and multi-substrate reasoning chains.
- Ramp photonic lattice emitters and alignment rings through the certified safe-state sequence.
- Maintain reactor protection on independent deterministic hardware.
- Avoid abrupt A.R.C. field changes near superconducting reactor magnets.

6.0 - DEGRADED OBSERVER MODES

Mode	Capability
A.R.C. Normal	Full photonic-neuromorphic operation, coherent memory, long-range temporal correlation
A.R.C. Degraded	Restricted substrate federation; shortened reasoning horizon; no autonomous infrastructure changes
Field Safe State	Conventional hardened compute only; read-only telemetry and safety advisory functions

7.0 - MAGNET AND CRYOGENIC RESPONSE

- Detect local resistive voltage and temperature rise.
- Open fast DC breakers and transfer stored magnetic energy to external dump resistors.
- Isolate affected cryogenic sectors and vent through rated relief manifolds.
- A generalized cryogenic flood is prohibited.
- Keep breeder-bay cryogenic loops isolated from primary-array loops wherever practical.

8.0 - DEC AND POWER-QUALITY FAULTS

- Open the affected core channel and discharge stored electrical energy into rated resistor banks.
- Preserve filtered power to the A.R.C. Field from redundant stores if stable.
- If phase-noise or timing quality exceeds certified limits, command A.R.C. degraded mode before coherence loss.
- Prevent raw converter or breeder-pulse transients from reaching AGI substrate buses.

9.0 - RESTART CRITERIA

- Correct core and breeder configuration verified: six primary, two auxiliary breeder.
- Pressure, vacuum, radiation, insulation, and isotope-accountancy tests passed.
- A.R.C. Field resonance map validated under representative substrate load.
- Cross-substrate memory-coherence and temporal-correlation tests passed.
- Human operations board and independent safety cell authorize restart.

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